

Figure 2: CouchDB Web interface

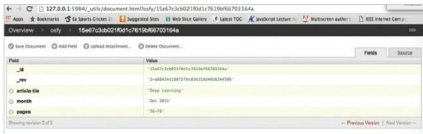


Figure 3: Adding documents and fields



Figure 4: MapReduce function

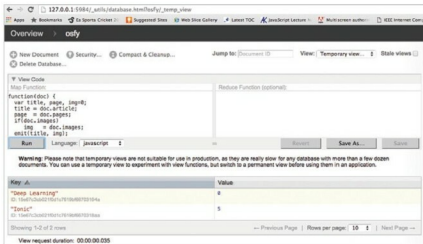


Figure 5: A sample map function

`curl -X PUT http://127.0.0.1:5984/osfy`

If the database is created successfully, then the response will be as follows:

```
{ "ok": true }
```

If you again list all the databases,

a new entry with the name 'osfy' can be noted:

```
[ "_replicator", "_users", "baseball", "mycontacts", "osfy", "test_suite_db", "test_suite_db2" ]
```

Futon

Futon is the Web based interface for

Apache CouchDB. It can be accessed with the URL http://127.0.0.1:5984/_utils/index.html, as illustrated in Figure 2.

The new documents in the database can be added by clicking on the 'New document' link. Fields can be added by clicking on 'Add Fields', as shown in Figure 3.

The JSON based source of the documents, with its fields, is shown below:

```
{
  "_id":
    "15e67c3cb021f0d1c7619bf66703164a",
  "_rev":
    "3-7121a64ddac1b604ee27681ff6b7a2a7",
  "month": "Dec 2015",
  "article title": "Deep Learning",
  "pages": "76-79"
}
```

Do note that it is not mandatory to have an identical field structure in all the documents. For example, a new field 'image count' has been added to the second document as shown below:

```
{
  "_id":
    "15e67c3cb021f0d1c7619bf6670318aa",
  "_rev":
    "1-b02bb3077cd66496142d4c38ed2353b7",
  "month": "Jan 2016",
  "article": "Ionic",
  "pages": "80-84",
  "image count": 5
}
```

In addition to adding textual fields, files can be uploaded as attachments. The attachments can be added by clicking the 'Upload Attachment' link in the document section.

CouchDB – Data retrieval

The retrieval of data is as important as storing it, if not more so. To retrieve data from the CouchDB store, MapReduce is used. As there is no predefined structure, firing of queries to filter out the required data is not expected here. The retrieval is done using the MapReduce functionality. The code can be written in JavaScript. There are other options such as CoffeeScript.